


Humidity

$$\rightarrow Y = \frac{y_A}{y_B} = \frac{p_A}{p_B} = \frac{p_A}{p_t - p_A} \quad \frac{\text{moles of A (Vapour)}}{\text{moles of B (gas)}}$$

$$\rightarrow Y'_{(\text{mass})} = Y \frac{\text{mass of A}}{\text{mass of B}} \left[\frac{\text{kg of water vap}}{\text{kg of dry air}} \right]$$

→ Saturation

↳ when partial pressure of water is equal to its vapour pressure present in air

Partial pressure = Vapour pressure

↓
we say air is completely saturated at particular temp

Saturated air

$$Y'_S = \frac{p_A^V}{p_t - p_A^V} \cdot \frac{M_{w, \text{water}}}{M_{w, \text{air}}}$$

$$p_A^V = A - \frac{B}{T - C} \quad (T \text{ is in } ^\circ\text{K})$$

↳ Antoine eqn.

∴ humidity = $\frac{Y'}{Y'_S} \times 100$ → gives idea how close is it to saturation

$$\therefore \text{relative humidity} = \frac{p_A}{p_A^V} \times 100$$

Humid volume

$$\text{m}^3/\text{kg of dry air} \leftarrow V_{tA} = \left(\frac{1}{M_{w, \text{air}}} + \frac{Y'}{M_w} \right) \times 22.14 \times \frac{T_G + 273}{273}$$

↗ dry air ↘ water vapour

humid heat $\rightarrow$ heat energy required to raise temp. of unit mass of air by 1°C
(water vap + dry air)

$$C_H = \underbrace{1.005}_{\text{heat capacity of dry air}} + \underbrace{1.884}_{\text{heat capacity of water vapour}} \text{ KJ/kg of dry air} \cdot \text{K}$$

Enthalpy

$$H' = \underbrace{C_H(T_{GT} - T_0)}_{\text{sensible heat}} + \lambda_H Y' \quad \text{latent heat}$$

$$= (1.005 + 1.884)(T_{GT} - T_0) + \underbrace{2260}_{\text{latent heat of vaporisation}} Y'$$

Dry bulb temperature

$$\text{WBT} \quad t_{G2} - t_w = \frac{h_s}{h_G / K_y} (y_w' - y')$$

$$\text{AST} \quad t_{G2} - t_{AS} = \frac{h_s}{C_H} (y_s' - y')$$

$$Le \rightarrow \frac{h_G / K_y}{C_H} \rightarrow \text{Lewis number}$$

$$\boxed{\frac{h_G}{K_y C_H} = Le^{0.569}}$$

at fully saturated condition (100% humidity)

$$T_{DBT} = T_{WB T} = T_{AST} = T_{DP}$$

→ For psychrometric chart you need two values, two get all 5 values

Applications of humidification operation

Major / important:

- Humidification of gases for controlled drying of wet solids
- Dehumidification and cooling of gas in air conditioning
- Gas cooling with the help of water
- Cooling of liquid (e.g. water) before reuse

(Absolute) humidity

It is the direct measurement of moisture content in a gas.

$$Y = \frac{y_A}{y_B} = \frac{\bar{p}_A}{\bar{p}_B} = \frac{\bar{p}_A}{p_t - \bar{p}_A} \frac{\text{moles A}}{\text{moles B}} \quad (\text{in terms of moles})$$

$$Y' = Y \frac{M_A}{M_B} = \frac{\bar{p}_A}{p_t - \bar{p}_A} \frac{M_A}{M_B} \frac{\text{mass A}}{\text{mass B}} \quad (\text{in terms of mass})$$

For air-water vap. system, A: water vap. and B: air

Humid volume

The humid volume, v_H is defined as the volume of unit mass of dry air with accompanying water vapor at a given temperature and pressure.

$$v_H = \left(\frac{1}{M_a} + \frac{Y'}{M_w} \right) 22.4 \frac{T_a + 273}{273} \frac{1.013 \times 10^5}{p_t}$$

where v_H is in m^3/kg

T_a is DBT of air in $^\circ\text{C}$

p_t is the pressure of air in Pa

For dry air: $Y' = 0$ and for saturated air $Y' = Y_s'$

Humid heat

The humid heat, c_H is the heat energy required to raise the temperature of unit mass of air with the accompanying water vapor by one (1) degree at constant pressure.

$$C_H = 1.005 + 1.88Y' \quad \text{kJ/(kg of dry air) } \cdot K$$

first part of right hand side is heat capacity of dry air in kJ/kg.K
and second part is heat capacity of water vapor in kJ/kg.K.

Enthalpy

The enthalpy of a vapor-gas mixture is the sum of the relative enthalpies of gas and vapor content.

$$H' = c_H(T_G - T_0) + \lambda_w Y'$$
$$= (1.005 + 1.88Y')(T_G - T_0) + 2260Y'$$

where λ_w is the latent heat of vaporization of water

H : kJ/kg of air-water vap. mixture

Difference of WBT and AST

Both of these are the temperatures when the rate of latent heat of vaporization extracted from bulk water becomes equal to the rate of sensible heat transfer from air to water.

- WBT is reached when there is large amount of air and small amount of water while AST is reached when small amount of air and larger amount of water.
- When WBT is reached the temperatures of water and air are different (temperature of water lower than that of air) while when AST is reached temperatures of water and air leaving are same.
- AST is either equal to or lower than WBT (equal in air-water system and unequal in other liquid-gas systems).
- When AST is reached air becomes saturated with water while when WBT is reached there is considerably no change in the humidity of air.
- AST is reached in adiabatic saturator while WBT is reached in wet wick thermometer.
- AST requires adiabatic conditions that heat is exchanged only between the passing air and water inside the adiabatic saturator while WBT requires heat exchange with surrounding air in open.
- AST is the maximum limit up to which temperature of water can be lowered by evaporation at any given dry bulb temperature and pressure in adiabatic conditions while WBT is the maximum limit up to which the temperature of water can be lowered by evaporation in open.
- AST is the equilibrium Temperature while the WBT is the steady state temperature.

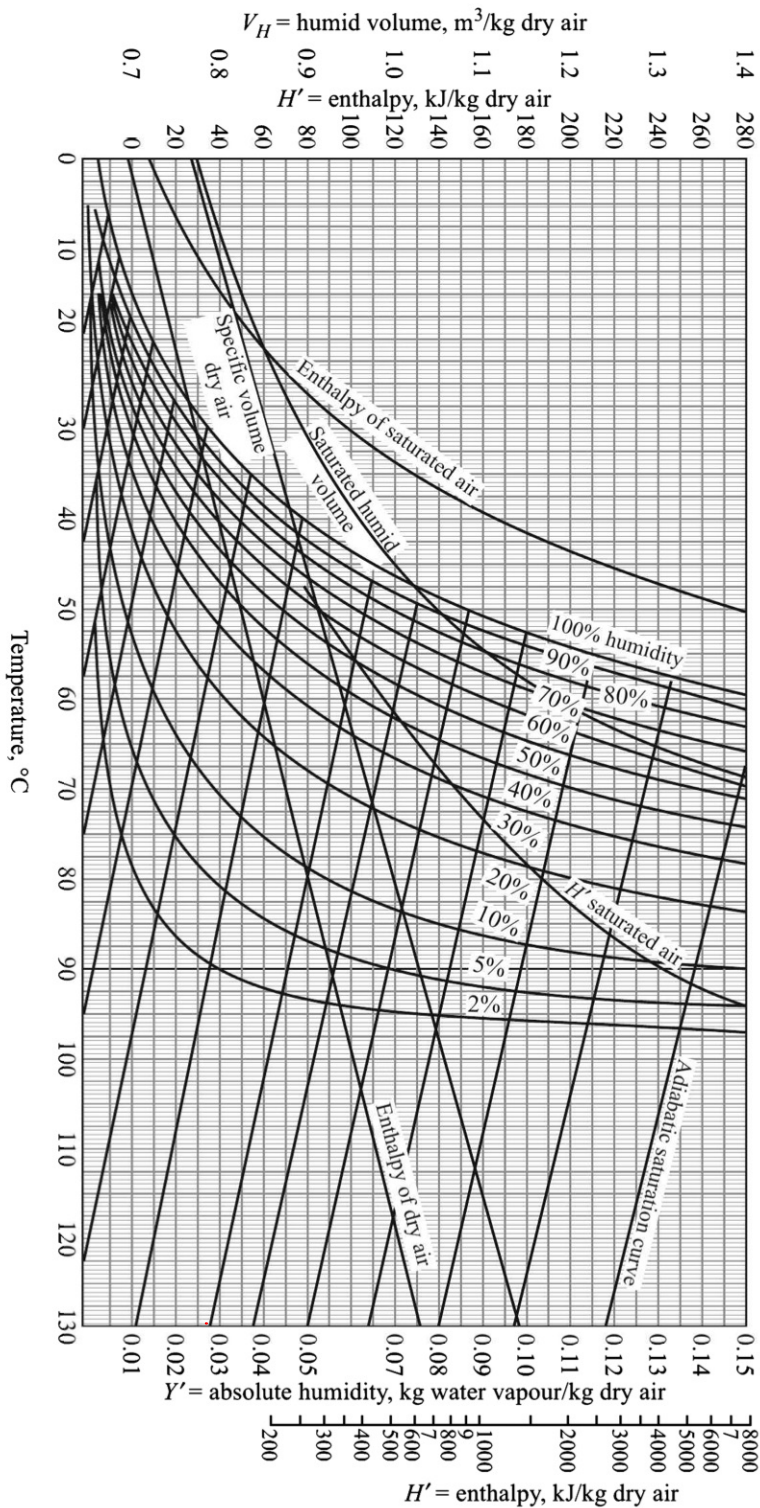
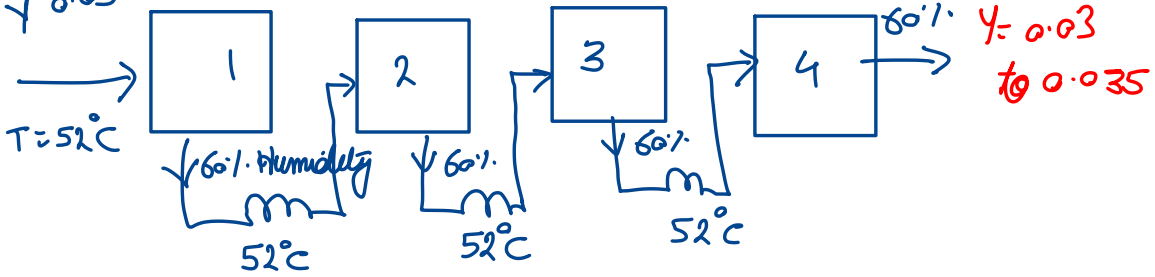


Figure 10.6 (a) Psychrometric chart for the air–water system at 1 atm total pressure.

20/10/22

$\gamma = 0.05$

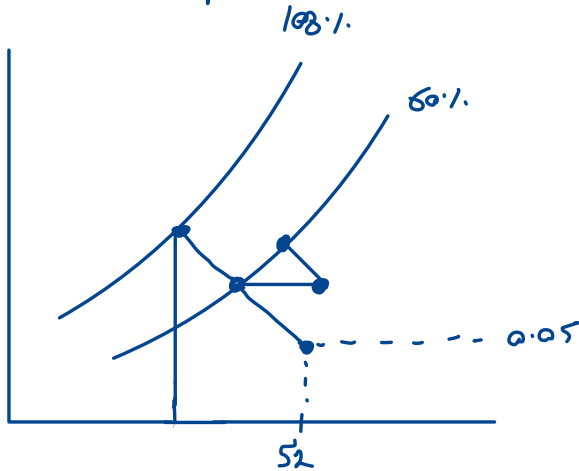


$\gamma = 0.03$
to 0.035

→ Moisture content is increasing, although the γ i.e. 60% remains same

Step

① locate the point, note the γ humidity value



Material is in equilibrium with air, so wet bulb temp of air = temp leaving the trays of material

$$\Delta\gamma = (0.03 - 0.005) \text{ kg wet vap / kg dry air}$$

Temp of tray 1 → 23°C
Temp of tray 2 → 30°C
Temp of tray 3 → 34°C
Temp of tray 4 → 35°C

V_H @ air leaving system (60%)

$$\left[\left(\frac{Q}{V_H} \right) \Delta\gamma \right] \text{ kg water vap / s}$$

Volumetric flow rate will be different at each stage,
the mass flow rate of dry air will be same throughout
↳ mass flow rate of dry air leaving
the system

25/10/22

Toluene air mixture @ 60°C

$$Y' = 0.05 \text{ kg vap / kg dry air}$$

$$\rho = 1.06 \text{ kg/m}^3$$

$$\text{Pressure} = 1 \text{ atm}$$

$$\text{Diffusivity of toluene} = 0.92 \times 10^{-5} \text{ m}^2/\text{s}$$

$$\text{viscosity} \quad \text{,,} \quad \text{,,} \quad = 0.0195 \text{ mPa}\cdot\text{s}$$

WBT? Adiabatic sat temp?

$$\left(\underset{\substack{\downarrow \\ 60^\circ\text{C}}}{T_{\text{DBT}} - T_{\text{WB}}} \right) = \frac{r}{(h_g / \text{K}_y)} (Y'_s - Y') \quad \rightarrow 0.05$$

$$Sc = \frac{\mu \beta}{D}$$

$$\left[Le \neq 1 \right. \\ \left. T_{\text{AST}} \neq T_{\text{WBT}} \right]$$

$Pr \rightarrow 0.7 - 0.8 \rightarrow$ in this range $\rightarrow$ Prandtl No.

$$Le \rightarrow \frac{Sc}{Pr} = \frac{2}{0.75} \rightarrow \text{Not close to 1} \rightarrow \text{Lewis no.}$$

If Le is close to 1, then
adiabatic sat temp = wet bulb
temp, otherwise adiabatic sat temp $\neq$ wet bulb temp.

$$\frac{(h_{G1}/K_Y)}{C_H} = e^{0.567}$$

$$\Rightarrow \frac{h_{G1}}{K_Y} = 1862.14796 \text{ J/kg}$$

$$\hookrightarrow 1005 + \underset{\substack{\uparrow \text{toluene} \\ 1256 \text{ J/kg}}}{C(0.05)}$$

$$\underbrace{(T_{DBT} - T_{WB})}_{\substack{\downarrow \text{known} \\ 60^\circ\text{C}}} = \frac{\underbrace{360 \text{ kJ/kg}}_{\substack{\uparrow \text{known}}} \underbrace{(Y'_s - Y')}_{\substack{\uparrow \text{unknown} \\ \hookrightarrow 0.05}}}{\underbrace{(h_{G1}/K_Y)}_{\substack{\uparrow \text{known}}}} \quad \text{--- (1)}$$

→ can't solve with T_{WB} , Y'_s with a single eqn, so need one more eqn

Antoine equation

$$\log_{10} p_s = A - \frac{B}{T_w + C} \quad \text{--- (2)}$$

$\nearrow \text{base} \rightarrow 4.1$
 $\nearrow 1346$
 $\nwarrow T_w + C \rightarrow 53$

$$Y'_s = \frac{p_s}{P - p_s} \frac{\text{mol wt of toluene}}{\text{mol wt of air}}$$

$\nwarrow \text{mol wt of toluene} \rightarrow 92$
 $\nwarrow \text{mol wt of air} \rightarrow 29$

→ Assume the initial value of WBT to be lower than the DBT
 $[WBT < DBT]$
 $\hookrightarrow$ because it is.

→ ① & ② are implicit, they can't be solved, so we need to do iterations here with an assumed initial assumption of $WBT < DBT$, then do iterations

Initial assumption $\rightarrow T_w \rightarrow 40^\circ C \rightarrow 313 K$

$$p_s = 0.0837$$

$$\gamma_s' = 0.2897$$

Note : Here $Le > 1$, so from Lewis no. only can use comment on whether $WBT > AST$ or $WBT < AST$

→ Ex. of Treybal

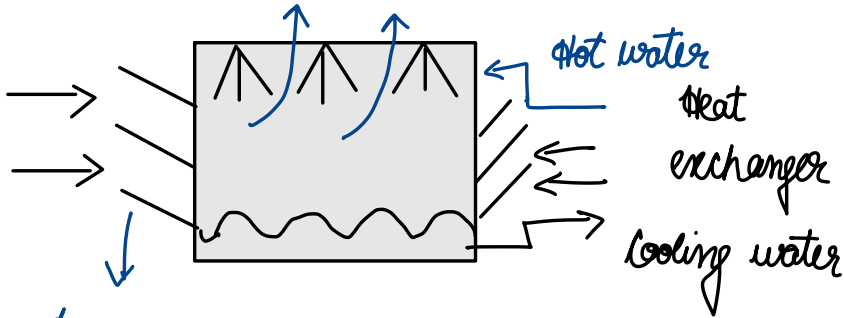
→ BK Dutta examples look

$\rightarrow WBT > AST$

$\rightarrow$ Sir said
reason find??

27/10/22

Cooling tower



Louvers
(like in train windows)

$$\text{Range} = T_{h,in} - T_{c,out}$$

↓
10-15°C

$$\text{Approach} = T_c - T_{WB,T} @ \text{ambient}$$

↗ room
↗ normal temperature
air

↘ ≥ 4°C

$$\text{Capacity} \equiv C \times R \times C_p [m\dot{C}_p \Delta T]$$

↓
cooling water
rate

↓
Range

↓
specific heat
capacity
of water

Basically heat
exchanged by the
cooling tower

If approach is less than
4° that means air contains
a good amount of humidity
and hence not good amount
of evaporation would be able
to happen, which would make
it inefficient

→ Factors affecting cooling towers operation are:-

- 1) Air pressure drop
2. Temp of warm water
3. DBT & WBT of air
4. Area of contact b/w water & air
5. Time of " " " "
6. Desired temperature drop

since then. A brief history of development of cooling towers has been given by Willa (1997).

The important factors that govern the operation of a cooling tower are: (i) the dry-bulb and wet-bulb temperatures of the air; (ii) the temperature of warm water; (iii) the efficiency of contact between air and water in terms of the volumetric mass transfer coefficient ($k_y \bar{a}$, see Section 10.7) and the contact time between the air and the water; (iv) the uniformity of distribution of the phases within the tower; (v) the air pressure drop; and (vi) the desired temperature of the cooled water. A tower is irrigated at the top through nozzles. A large air-water contact area and contact time are offered by a typical tower packing. Air may enter the tower

effectiveness

better use this

$$E = \frac{R}{R+A} = \frac{T_h - T_c}{T_h - T_{WB,T}}$$

→ If approach is high, it would be better.

performance of cooling tower will increase

→ Interplay of approach and range

→ We try to increase both effectiveness and approach try to balance.

Both effectiveness and approach should ideally be higher

Types of air flow pattern

→ big rectangular chamber
→ air is driven by its own velocity

→ atmospheric

→ Natural draft

→ Mechanical draft

→ Forced

→ Induced.

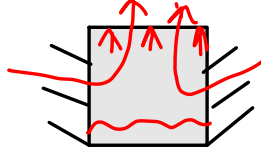
→ Cheap but inefficient

→ no control on flow

atmospheric → performance depends on direction & velocity of air

Rarely used

Natural draft



Area less

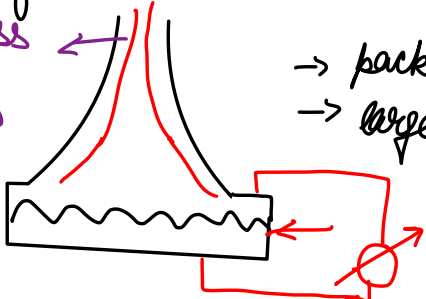
↑, ↓, ΔP will

drive the air flow from bottom to top

→ packing

→ large construction

→ increased intimacy



100m

natural draft tower.

Natural draft towers are used in big installations like steam power plants requiring a huge quantity of cooling water. The diameter may be as large as 100 m and the height about 150 m and the capacity may be as high as 500,000 gpm (gallons per minute) water. The hyperbolic shape is made because of the following reasons: (i) more packing can be fitted in the bigger area at the bottom of the shell; (ii) the entering air gets smoothly directed towards the centre because of the shape of the wall, producing a strong upward draft; and (iii) greater structural strength and stability of the shell is provided by this shape. The initial cost of a hyperbolic tower is high. However, the absence of a fan reduces the fixed cost and particularly the operating and maintenance costs making it economic over a long period of time.

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the forced draft system are the following: (i) A part of the velocity head of air thrown by the blower is converted to pressure head on entering into the tower. This makes the forced draft tower slightly more energy efficient than the induced draft type. (ii) The system is less susceptible to vibrations because the fans are installed near the ground on a solid foundation. The disadvantages of the forced draft system are: (i) the air flow through the packing or fill may not be uniform, particularly if the tower is big and a big fan is used; (ii) some of the warm and humid air may be recirculated back into the tower thereby reducing the performance (see Section 10.6.1). The recirculation rate becomes low if the wind velocity is high. Forced draft towers are not very popular except for small capacities.

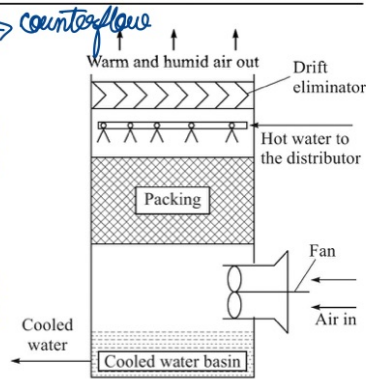


Figure 10.9 Schematic of a forced draft cooling tower.

Induced draft towers → can be both counter or cross flow
An induced draft tower has one or more fans located at the top. The fans suck fresh air through the air inlets at the bottom. The air flow or draft is 'induced' by the suction created by the fans and hence the name. The induced draft tower may be of two types—crossflow and counterflow—depending upon the air inlet and flow pattern. In a 'counterflow induced draft tower' (Figure 10.10), a vertical movement of air counter-current to the warm water occurs. The major advantage of this flow configuration is that the relatively dry air contacts the coldest water at the bottom, and the humid air contacts the warm water at the top of the tower. This ensures maximum average driving force for both heat and mass transfer. But more fan horsepower is required because of the restricted area of air flow at the bottom and, unlike the forced draft system, the velocity head of the air thrown out of the tower by the fan is dissipated.

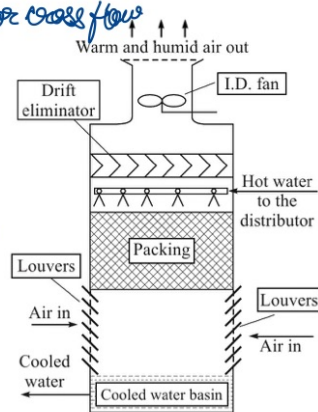
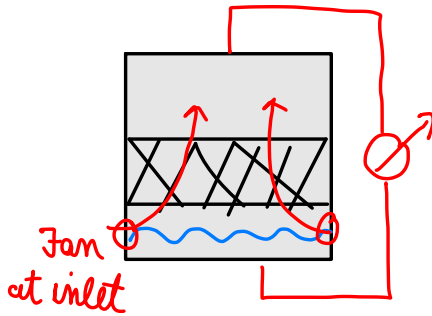


Figure 10.10 Schematic of a counterflow induced draft cooling tower.

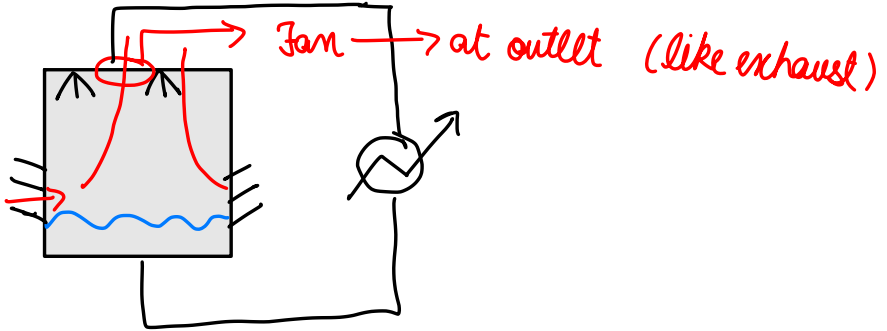
The crossflow induced draft cooling tower provides horizontal air flow along the packed height (Figure 10.11). The air flow is crosscurrent to the down-flowing water. Louvers are provided all along the walls of the tower to allow the air to move in. For the same air flow rate, the tower requires less motor horsepower than the counterflow type. But the growth of algae on the fills is more because the tower interior gets more sunlight through the larger number of louvers.

Forced

More cost in forced and induced



Induced



Cooling tower operation

source for loss of water

① Evaporation

Empirical relation ← For every 6° degree drop in water temp,
0.85% of circulation / water rate will be lost

$$\text{Evaporation rate} = 0.0085 \times \left(\frac{R}{6}\right) \times (\text{cooling rate})$$

$$Q = (kW) \times R \times C_p = (E_{\text{vap}}) \times R \times \text{latent heat of vaporisation}$$

② Entrainment / Drift loss → mist / water droplets carried
by air
In range of 0.2 - 1.1.

Blowdown → removal of part of concentrated water
and replacement with fresh, makeup water
↳ not continuous, after certain time period

11/11/22

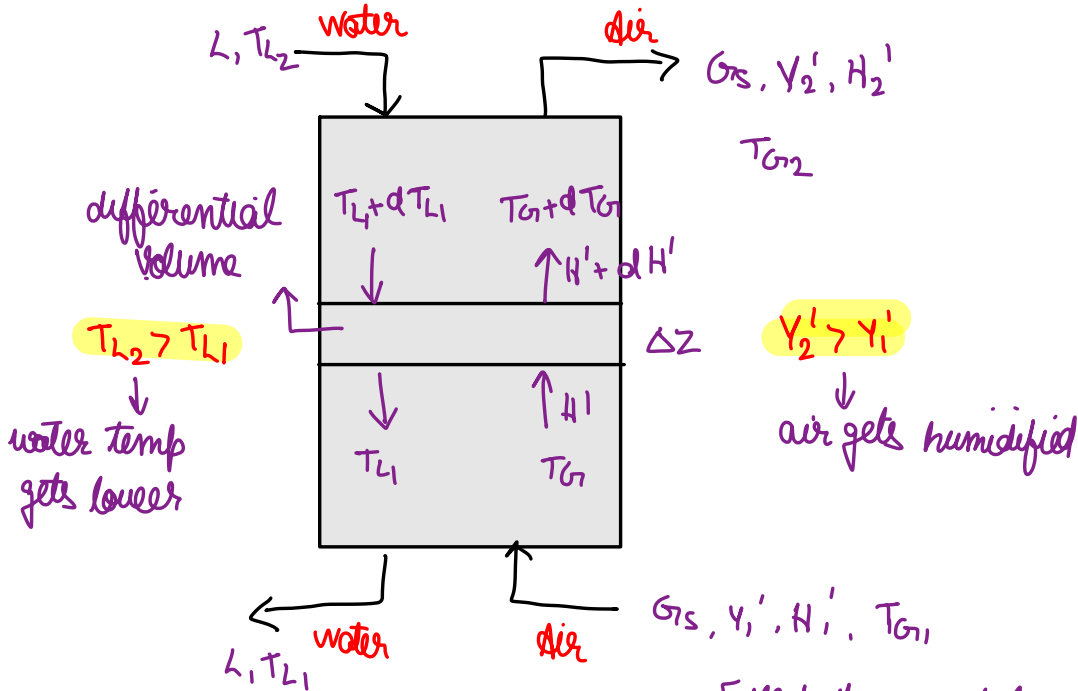
Process design of cooling tower

Assumptions

1) Adiabatic operation

2) Rate of water evaporation is small compared to circulation rate

water flow rate L is constant
(also we are adding make up water)



[all bottom symbols have subscript 1]

↓
not outlet, inlet

→ We have followed this convention

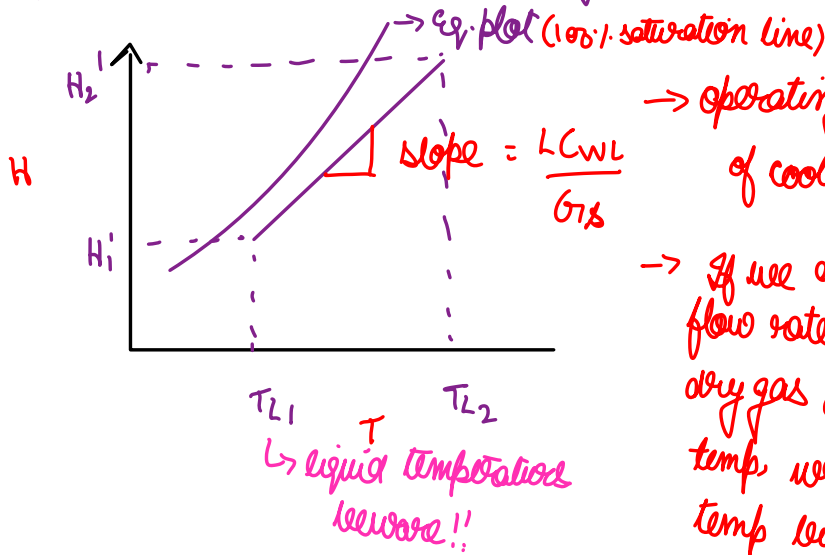
Enthalpy

$$\text{liq. flow rate} \leftarrow L C_{WL} dT_L = G_{s,d} dH' \rightarrow \text{enthalpy of air}$$

$\downarrow$ sp. heat cap of water $\downarrow$ liq. temp $\downarrow$ gas (dry gas) flow rate

$$L C_W (T_L - T_{L1}) = G_{s,d} (H' - H'_1)$$

$\swarrow$ at any arbitrary pt $\downarrow$ at inlet $\downarrow$ at inlet $\searrow$ at any arbitrary pt



$\rightarrow$ operating line of cooling tower

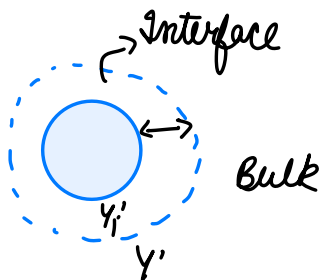
$\rightarrow$ If we are given flow rate of liquid, dry gas and inlet temp, we can get outlet temp but can't get height of the tower

Interfacial mass transfer

$$G_{s,d} dY' = K_y \bar{a} (Y_i' - Y') dz$$

$\hookrightarrow$ interfacial absolute humidity $\rightarrow$ assume it to be equal to saturated humidity

$\swarrow$ mass transfer coefficient $\hookrightarrow$ interfacial area of contact



Interfacial heat transfer

$$-G_S C_H dT_G = h_G \bar{a} (T_G - T_i) dz$$

-ve because $\swarrow$ interfacial temp is less than bulk
 $\downarrow$ heat transfer coefficient
 $\downarrow$ interfacial temp.

$$H' = \Gamma y' + C_H (T_G - T_0)$$

$\hookrightarrow$ heat of vaporisation
 $\hookrightarrow$ sensible heat

$\times G_S$

$$G_S H' = G_S \Gamma y' + C_H (T_G - T_0) G_S$$

$$G_S dH' = G_S \Gamma dy' + C_H G_S dT_G$$

using interfacial mass & heat transfer eqns.

$$G_S dH' = \Gamma K_y \bar{a} (y_i' - y') dz - h_G \bar{a} (T_G - T_i) dz$$

$$\therefore K_y \bar{a} dz \left(\Gamma (y_i' - y') - (C_H) (T_G - T_i) \right)$$

$\frac{h_G / K_y}{C_H} = h_e^{0.567} \rightarrow$ for air-water $h_e = 1$; $h_G / K_y = C_H$

$$G_s dH' = K_y \bar{a} dz (H_i' - H')$$

$$\int_{H_i'}^{H_2'} \frac{dH'}{H_i' - H'} = \frac{K_y \bar{a}}{G_s} \int_0^Z dz$$

→ will give height of cooling tower

↗ height of tower

$$H_{tG} = \frac{G_s}{K_y \bar{a}} \quad N_{tG} = \int_{H_i'}^{H_2'} \frac{dH'}{H_i' - H'}$$

$$Z = N_{tG} H_{tG}$$

H_i' → would be on eq. line

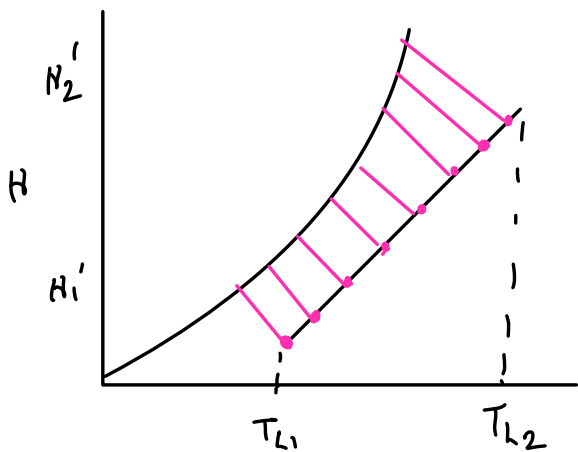
H' → would be on operating line

H'	H_i'	$H_i' - H'$	$1/(H_i' - H')$
$\vdots$			$\vdots$
$\vdots$			$\vdots$

$\underbrace{h_c dT}_{\text{liquid side heat transfer}} = h_L \bar{a} (T_L - T_i) dz \rightarrow \text{liquid side heat transfer}$

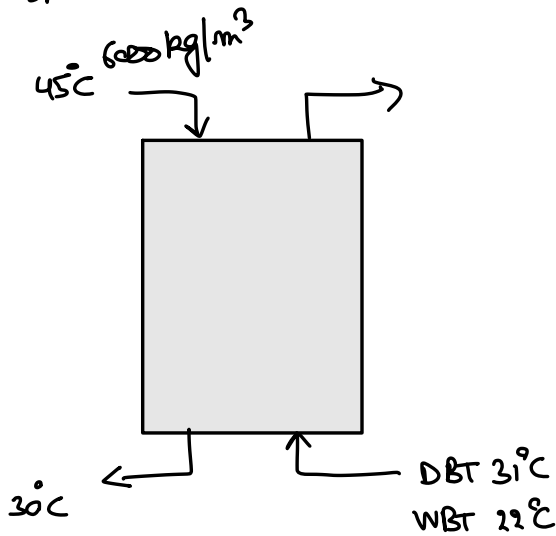
$$\hookrightarrow G_s dH \rightarrow G_s dH = K_y \bar{a} dz (H_i' - H')$$

$$\frac{H_i' - H'}{T_i - T} = - \frac{h_L}{K_y}$$



T_L	H'	H_i'	$H_i' - H'$	$1 / (H_i' - H')$
.				.
.				.
.				.

3/11/22



To get minimum flow rate of air $\Rightarrow$ Draw tangent to eq. curve and 1.4 times that will give the flow rate of air

$T^\circ(\text{C})$	p_v	$\Rightarrow$ saturation y_s humidity	C_H	H_s
20	0.0230	0.01442	1.03210	64.584
25	0.0313	0.01979	1.0422	77.0337
30	0.04196	0.0268	1.0553	93.284
35	0.0556	0.0360	1.0726	114.613
40	0.0730	0.04820	1.0956	142.896
45	0.0949	0.06416	1.1256	179.895
50	0.122	0.0850	1.1648	228.208

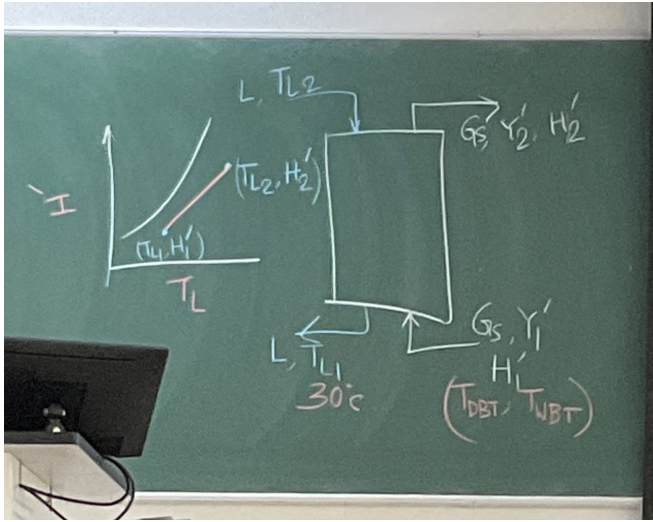
$$y_s = \frac{p_v}{p_T - p_v} \times \frac{M_w}{M_{air}}$$

$$p_T \Rightarrow 1 \text{ atm} - 1.013 \text{ bar}$$

$$M_w / M_{air} = 18/29$$

$$H_s' = 2264s' + \underbrace{(1.005 + 1.884s')}_{CH} \Delta T \rightarrow (T_G - T_o) \rightarrow \text{reference temp} = 0^\circ\text{C}$$

dry bulb temp



Adsorption

Langmuir isotherm

- 1) molecules adsorbed at discrete active sites
- 2) surface has equal energy over the whole surface
- 3) No interaction b/w molecules
- 4) Each site adsorbs one molecule.

$$\theta = \frac{KP}{1+KP}$$

$$K \rightarrow \frac{a_0 \rightarrow \text{adsorption const}}{d_0 \rightarrow \text{desorption const}}$$

$$a_0 P S_0 = a_0 S_1$$

free $\leftarrow$ $\rightarrow$ occupied

Freundlich Isotherm

$$q_s = K P^{1/n}$$

$\rightarrow$ for non-uniform energy of surface $\rightarrow$ so heat of adsorption is different for each site.

$\rightarrow$ Fails at high pressures

$\rightarrow$ BET isotherm

Assumptions

- $\rightarrow$ Multilayers
- $\rightarrow$ No transmigration
- $\rightarrow$ Equal energy
- $\rightarrow$ Langmuir to each layer

